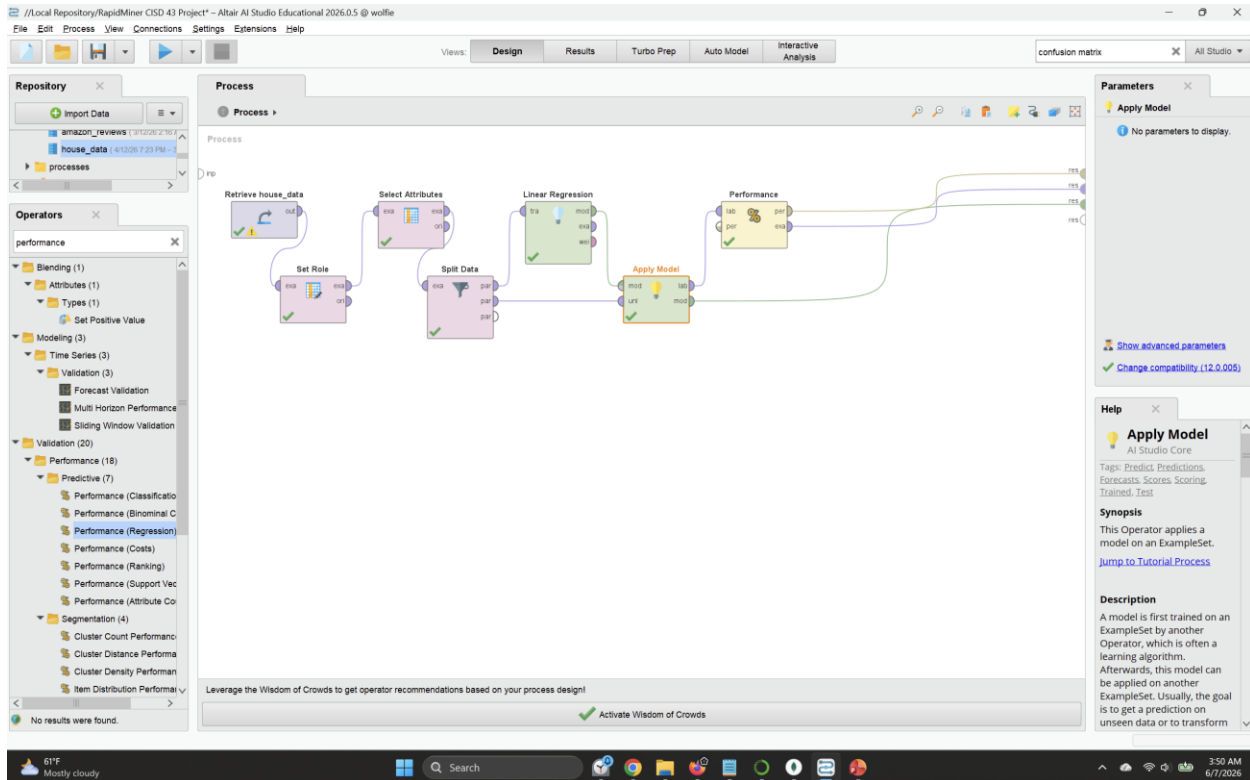


RapidMiner Data Analysis, Tests, and Visualizations

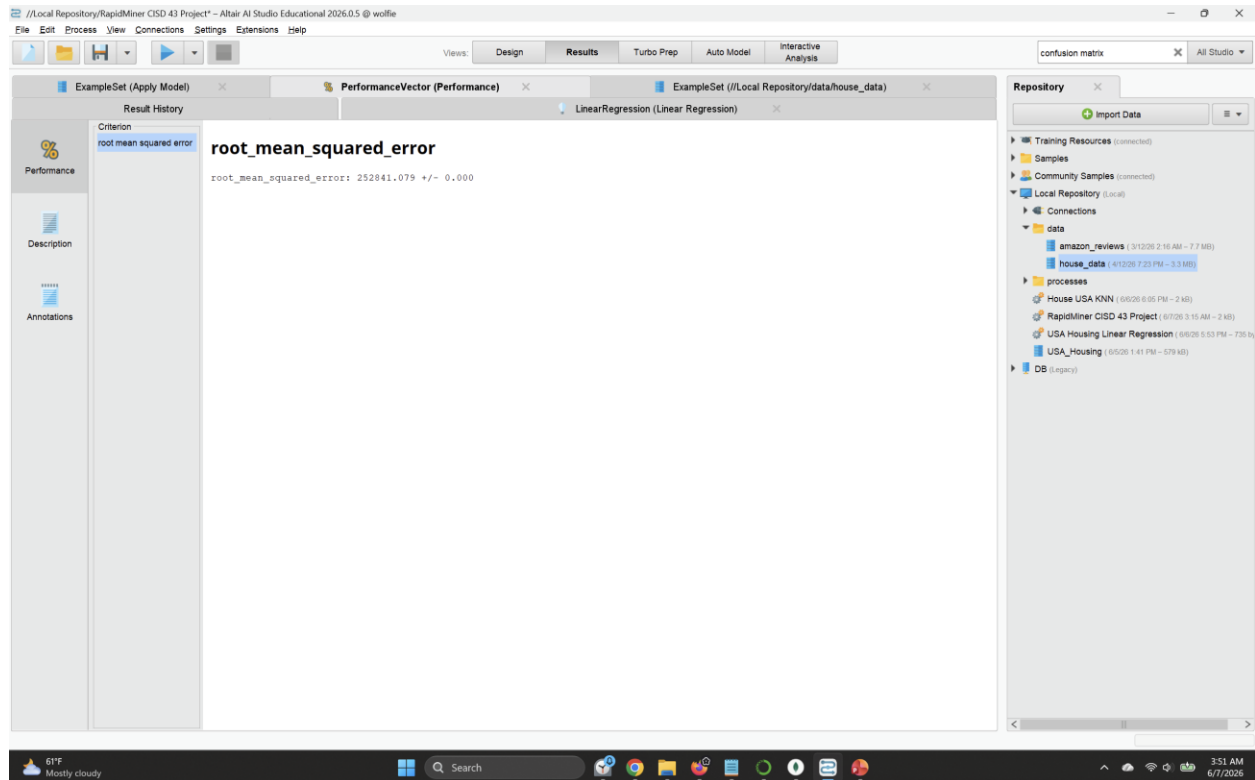
Note:

The free license version of RapidMiner has data limits, so the dataset here is truncated to 5,000 records.

TEST 1: Linear Regression:

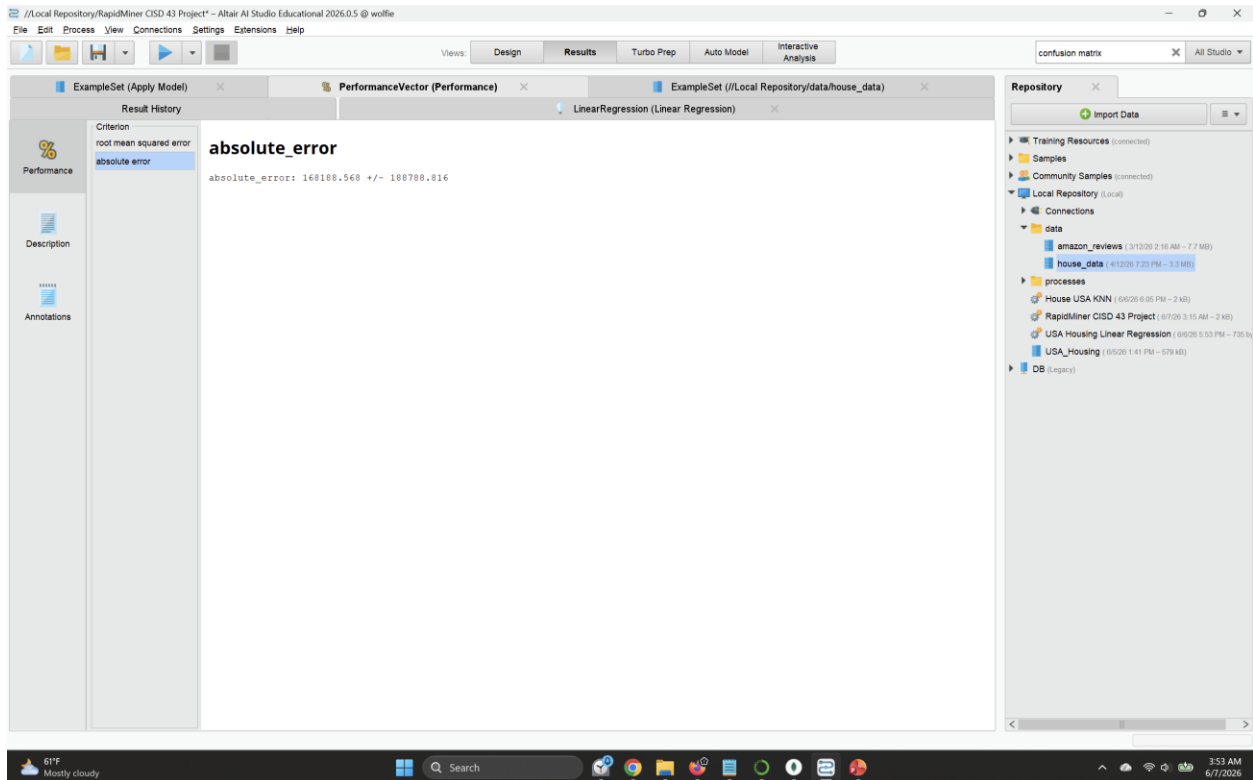


Here is the the model's root mean squared error:



Root mean squared error (RMSE) measures the difference between predicted values and actual values. The lower the RMSE, the more accurate the model.

Here is the absolute error:



Absolute error measures how far the estimated value is from the actual value. The lower the absolute error, the better the model.

Here are breakdown statistics by each predictor variable:

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File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

confusion matrix X All Studio

ExampleSet (Apply Model) PerformanceVector (Performance) ExampleSet (/Local Repository/data/house_data)

Result History LinearRegression (Linear Regression)

Attribute	Coefficient	Std. Error	Std. Coefficient	Tolerance	t-Stat	p-Value	Code
bedrooms	-56494.364	2603.638	-0.144	0.655	-21.698	0	****
bathrooms	7075.172	3922.818	0.015	0.425	1.804	0.071	*
sqr_living	309.583	3.454	0.771	0.388	89.628	0	****
(Intercept)	71401.077	7746.288	?	?	9.217	0	****

Repository

Import Data

- Training Resources (connected)
- Samples
- Community Samples (connected)
- Local Repository (local)
 - Connections
 - data
 - amazon_reviews (3/12/26 2:16 AM - 7.7 MB)
 - house_data (4/12/26 7:23 PM - 3.3 MB)
 - processes
 - House USA KNN (6/6/26 6:05 PM - 2 KB)
 - RapidMiner CISO 43 Project (6/7/26 3:15 AM - 2 KB)
 - USA Housing Linear Regression (6/6/26 5:53 PM - 736 B)
 - USA_Housing (6/5/26 1:41 PM - 579 KB)
 - DB (Legacy)

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Steps

First, I imported the dataset into RapidMiner, in the local repository.

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Views: Design Results Turbo Prep Auto Model Interactive Analysis

confusion matrix X All Studio

Result History

ExampleSet (Retrieve house_data) X ExampleSet (//Local Repository/data/house_data) X

Open in Turbo Prep Auto Model Interactive Analysis

Filter (21,613 / 21,613 examples): all

Row No.	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition
1	7129300520	20141013T0...	221900	3	1	1180	5650	1	0	0	3
2	6414100192	20141209T0...	538000	3	2.250	2570	7242	2	0	0	3
3	5631500400	20150225T0...	180000	2	1	770	10000	1	0	0	3
4	2487200875	20141209T0...	604000	4	3	1960	5000	1	0	0	5
5	1954400510	20150218T0...	510000	3	2	1680	8080	1	0	0	3
6	7237550310	20140512T0...	1225000	4	4.500	5420	101930	1	0	0	3
7	1321400060	20140627T0...	257500	3	2.250	1715	6819	2	0	0	3
8	2008000270	20150115T0...	291850	3	1.500	1060	9711	1	0	0	3
9	2414600126	20150415T0...	229500	3	1	1780	7470	1	0	0	3
10	3793500160	20150312T0...	323000	3	2.500	1890	6560	2	0	0	3
11	1736800520	20150403T0...	662500	3	2.500	3560	9796	1	0	0	3
12	9212900260	20140527T0...	468000	2	1	1160	6000	1	0	0	4
13	114101516	20140528T0...	310000	3	1	1430	19901	1.500	0	0	4
14	6054650070	20141007T0...	400000	3	1.750	1370	9680	1	0	0	4
15	1175000570	20150312T0...	530000	5	2	1810	4850	1.500	0	0	3
16	9297300055	20150124T0...	650000	4	3	2950	5000	2	0	3	3
17	1875500060	20140731T0...	395000	3	2	1890	14040	2	0	0	3
18	6865200140	20140529T0...	485000	4	1	1600	4300	1.500	0	0	4
19	16000397	20141205T0...	189000	2	1	1200	9850	1	0	0	4
20	7983200060	20150424T0...	230000	3	1	1250	9774	1	0	0	4
21	6300500875	20140514T0...	385000	4	1.750	1620	4980	1	0	0	4
22	2524049179	20140826T0...	2000000	3	2.750	3050	44867	1	0	4	3
23	7137970340	20140703T0...	285000	5	2.500	2270	6300	2	0	0	3
24	8091400200	20140516T0...	252700	2	1.500	1070	9643	1	0	0	3

ExampleSet (21,613 examples, 0 special attributes, 21 regular attributes)

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Connections

data

amazon_reviews (3/12/26 2:16 AM - 7.7 MB)

house_data (4/12/26 7:23 PM - 3.3 MB)

processes

House USA KNN (6/6/26 6:05 PM - 2 KB)

USA Housing Linear Regression (6/6/26 5:53 PM - 736 B)

USA_Housing (6/5/26 1:41 PM - 579 KB)

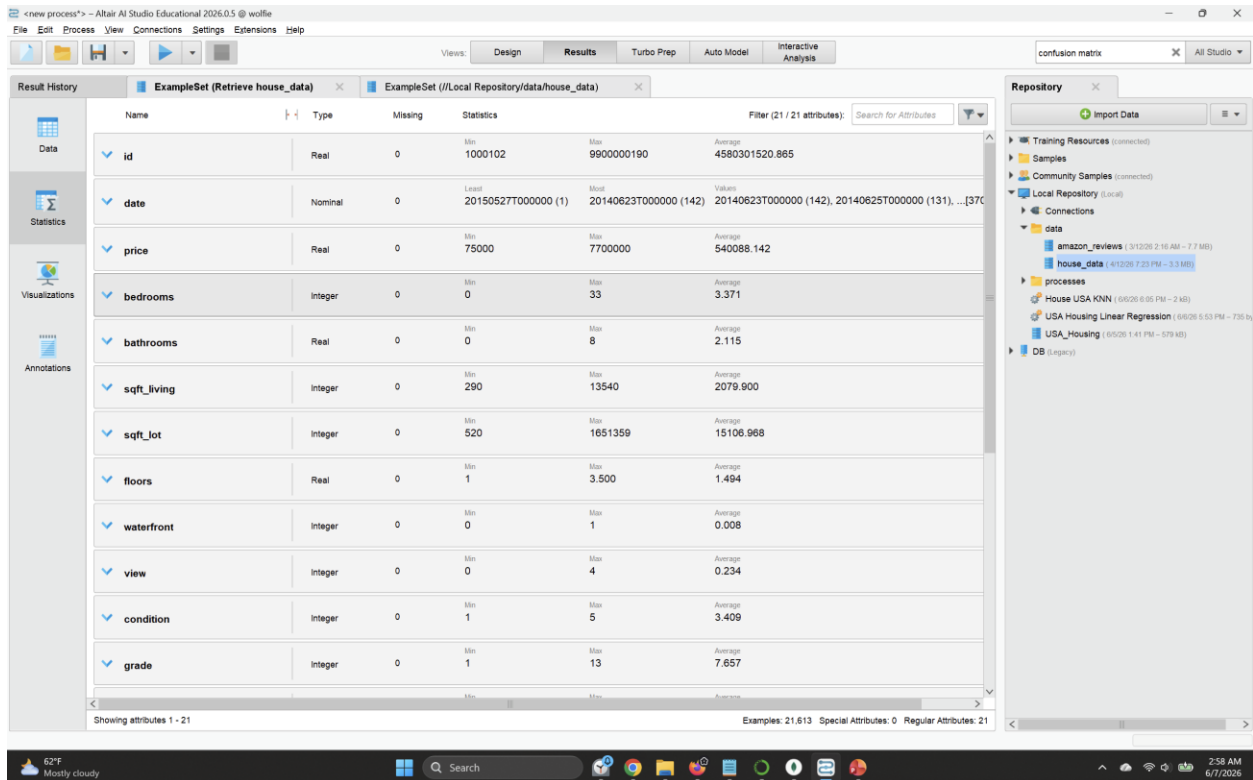
DB (Legacy)

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Search

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Next, I checked if any data cleaning was necessary due to missing or inappropriate entries. The third column below, “Missing”, shows that there are no missing or improperly formatted values for any of the dataset’s attributes.



1: Linear Regression Test (detailed steps for my future reference)

Here I add a process path in the Processes window via dragging and dropping. It is a CSV file; its separator value does not need to be changed.

As I set up the process, I connect all of the nodes to make a process path. Drop the following into the Process Window:

--Data: drop the data into the Process window

--Set Role (mousing over the Set Role box also gives you the datatype of your variables and the number of missing values). Double click on this process box

-->For a Linear Regression that predicts the value of a house, set the attribute as Price and the target role as “label” (and then click “Apply”).

*To verify the intended variable was selected, click Play. The Price variable should be highlighted.

--Set Attributes: Select attribute type: a subset

I selected “bedrooms”, “bathrooms”, and “square feet of living area”.

--Split Data: The model’s fit needs to be tested by splitting the data into Training data and Testing data. I chose an 80/20 ratio of training/testing data.

--Linear Regression (machine learning type): add the Linear Regression node to the Process

--Apply model: this node will apply the model. Apply the training data (top output of box) to the Apply Model node. The test data will not be run through the model though, so connect that straight to apply model and bypass Linear Regression

--Performance: in this case, select “Performance (Regression)”, since this instance of machine learning will be a linear regression. Connect to the end terminus of the entire process, and also connect the lower output of “Apply Model” to the terminus as well. Next, connect the lower output bubble of the Performance to the end of the process path to complete the entire process.

--In the Parameters panel on the right, select tests. Here I choose the Root Mean Square Error (RMSE) and the Absolute Error.

--Execute the process by pressing the large Play button on the upper left.

ANALYSIS 2: Generate a Correlation Matrix

I used the Correlation Matrix node to generate a correlation matrix. I selected only the attributes I used above in the Linear Regression model, namely Price, Number of Bedrooms, Number of Bathrooms, and Square Feet of Living Space.

The screenshot displays the Altair AI Studio interface. The main workspace shows a process flow with three nodes: 'Retrieve house_data', 'Select Attributes', and 'Correlation Matrix'. The 'Retrieve house_data' node is connected to 'Select Attributes', which is then connected to 'Correlation Matrix'. The 'Correlation Matrix' node has two output ports, one of which is connected to a terminus. The 'Parameters' panel on the right is open for the 'Select Attributes' node, showing settings for 'type' (inc...), 'attribute filter type' (a s...), and 'select subset' (Se...). The 'Help' panel on the right provides information about the 'Select Attributes' operator, including its synopsis and description. The bottom status bar shows the system temperature as 61°F and the date as 6/7/2026.

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File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

confusion matrix X All Studio

Result History

Correlation Matrix (Correlation Matrix) X

ExampleSet (/Local Repository/data/house_data) X

Data

Pairwise Table

Matrix Visualization

Plot view

Annotations

Attribut...	price	bedroo...	bathroo...	sqft_livi...
price	1	0.308	0.525	0.702
bedrooms	0.308	1	0.516	0.577
bathrooms	0.525	0.516	1	0.755
sqft_living	0.702	0.577	0.755	1

Repository

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Samples

Community Samples (connected)

Local Repository (Local)

Connections

data

amazon_reviews (3/12/26 2:16 AM - 7.7 MB)

house_data (4/12/26 7:23 PM - 3.3 MB)

processes

CISD 43 Capstone Project Correlation Matrix (6/7/26 4:14 AM - 2 KB)

House USA KNN (6/6/26 6:05 PM - 2 KB)

RapidMiner CISD 43 Project (6/7/26 4:05 AM - 5 KB)

USA Housing Linear Regression (6/6/26 5:53 PM - 735 bytes)

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